

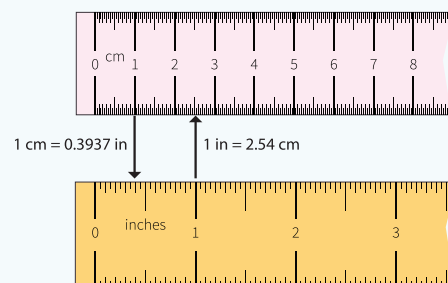
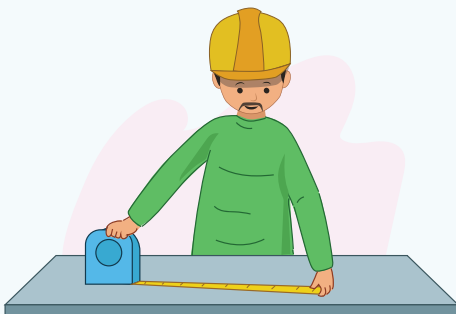
# 3D Furniture Design



Measurement is vital to learn about the size of an object. Let's learn how we can use units to measure spaces and objects.



Sometimes, furniture in a room is put together as a part of a larger theme. A theme adds purpose, style, and personality to a space. Objects in our surroundings are designed in a particular manner to address specific requirements. The space and furniture need to be measured correctly to make sure the area is used well.



## 2. 3D Furniture Design

A distance, however long or short, can be measured in different units. A millimetre is the smallest unit of measurement for length and distance. We can convert a distance or length measurement from one unit to another using formulas.

$$1 \text{ inch (in)} = 0.0833333333333334 \text{ foot (ft)}$$

$$1 \text{ foot (ft)} = 12 \text{ inches (in)}$$

$$1 \text{ metre} = 100 \text{ centimetres}$$

$$1 \text{ metre} = 1000 \text{ millimetres}$$

There are two main units of measurement:

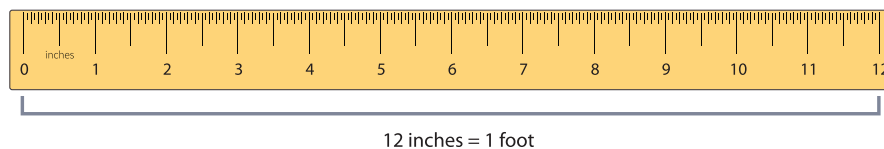
- a. U.S./Imperial units- (U.S. is expanded as United States)
- b. SI or Metric units- (SI is expanded as The International System of Units)

Inch and foot are units of length in the Imperial or the US system.

| Type   | U.S. Unit                  | Metric Unit                                  |
|--------|----------------------------|----------------------------------------------|
| Length | Inches, Feet, Yards, Miles | Millimetres, Centimetres, Metres, Kilometres |
| Weight | Ounces, Pounds             | Grams, Kilograms                             |
| Time   | Seconds, Minutes, Hours    | Seconds, Minutes, Hours                      |
| Volume | Ounces, Gallons            | Millilitres, Litres                          |

### Difference between inch and foot

There are 12 inches in 1 foot. It means that a **foot** is greater than an **inch**.



In this session, we will use inches and feet as units of measurement.

#### Note:

The SI base unit for length is *metre*.

1 metre is equal to 39.3700787402 inches, or 3.28083989501 feet.

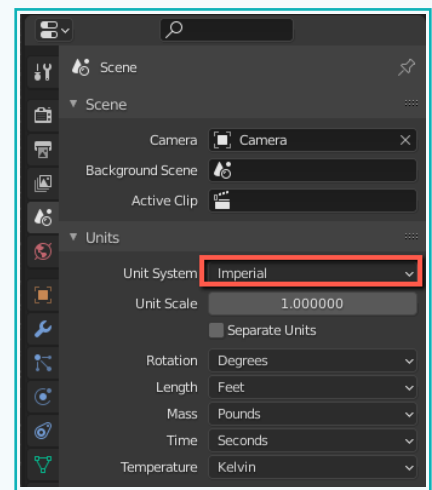
### Activity

Create a table in Blender using the imperial system.

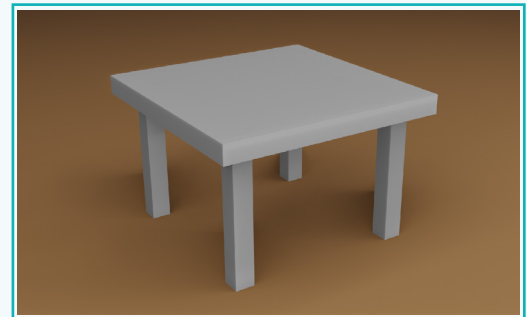
The dimensions of the table are as follows:

- Height (Z-value): 10 inches (10")
- Length (X-value): 1 foot (1')
- Width (Y-value): 1 foot (1')

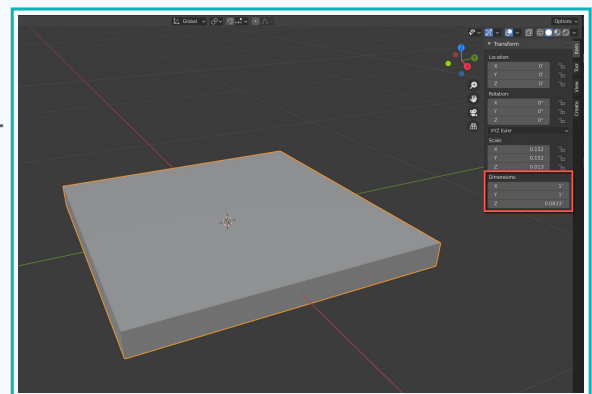
1 By default, the unit system that Blender uses is Metric. Since the dimensions provided are in imperial units, we need to modify it. In Scene > Units > Unit System, select Imperial.



2 Once the unit system is set, we need to understand the components required to create the table model.



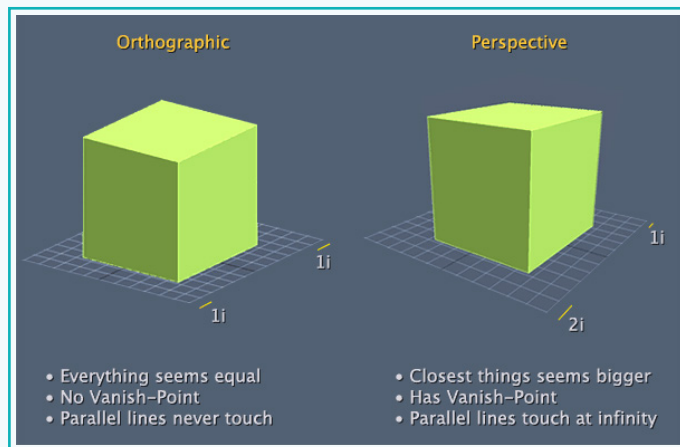
3 Start with the table top. For the top, we need a mesh-cube.



**Note:** 1 inch = 0.0833 ft.

## Activity

Observe how a 3D object looks. The maximum surface area of a 3D object is visible in perspective view and orthographic view.



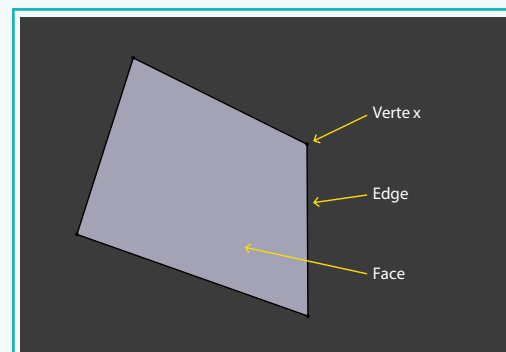
In Blender, we can toggle between orthographic and perspective view with the number 5 key on the numpad.

Similarly, different views can be easily accessed.

|                    |                         |                      |
|--------------------|-------------------------|----------------------|
| <b>7</b><br>Top    | <b>8</b><br>Up          | <b>9</b><br>Opposite |
| <b>4</b><br>Left   | <b>5</b><br>Persp/Ortho | <b>6</b><br>Right    |
| <b>1</b><br>Front  | <b>2</b><br>Down        | <b>3</b><br>Side     |
| <b>0</b><br>Camera |                         | <b>.</b><br>Focus    |

To create the legs for the table, we need to extrude sections of an object to create a composite shape. This allows us to work with the table as a single object, instead of a group of individual 3D shapes. This also allows us to work with a limited number of objects in large scale projects to make the work manageable.

Before we work on the sections of the object, let us understand what a 2D mesh object is composed of. With a mesh, everything is built from three basic structures: vertices, edges and faces.



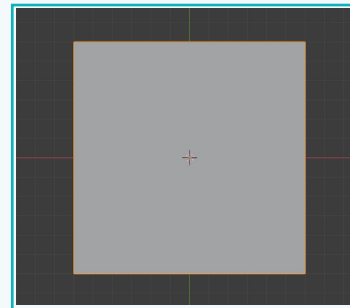
### Activity

The most elementary part of a mesh is the vertex (plural - vertices) which is a single point or a position in 3D space. A vertex is represented in the 3D Viewport in Edit Mode as a small dot. The vertices of an object are stored as an array of coordinates.

An edge is a straight line that always connects two vertices. The edges are the “wires” we see when we look at a mesh in wireframe view. They are usually invisible on a rendered image. Edges are used to construct faces.

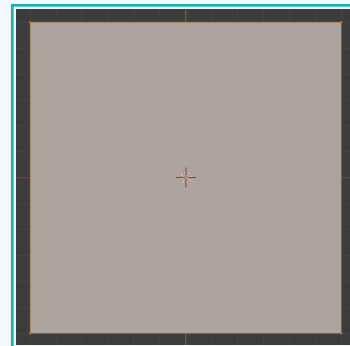
Faces are used to build the actual surface of an object. They are what we see when we render the mesh. If this area does not contain a face, it will simply be transparent or non-existent in the rendered image.

a. Switch to top view (press 7).

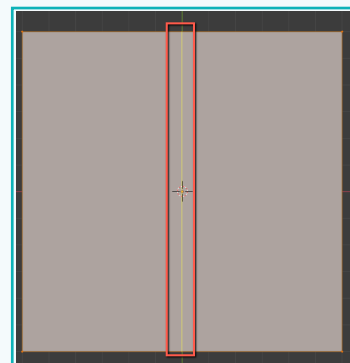


b. Change to edit mode (using the Tab key).

Edit Mode is the main mode where modeling takes place. The Edit Mode is used to edit or modify the primitive mesh.

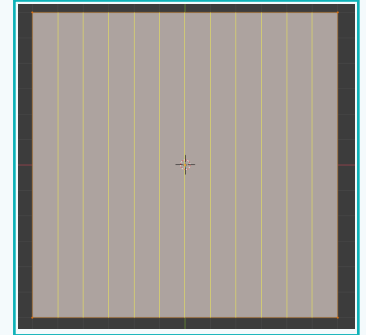


- 4 To divide the mesh, add edge loops using Ctrl + R. To do this, press Ctrl+R. A vertical yellow line will appear on the cube (showing the division of the cube into two vertical segments).

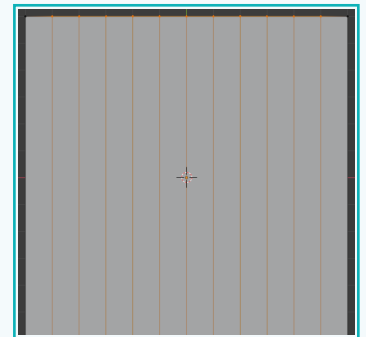


### Activity

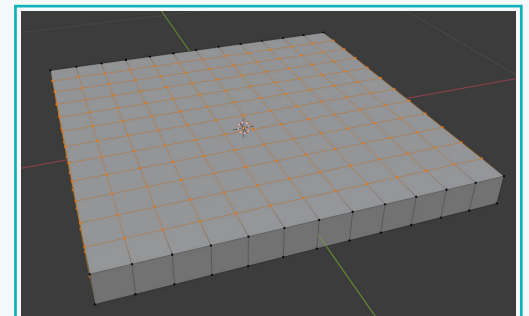
5 To add multiple edge loops, scroll up



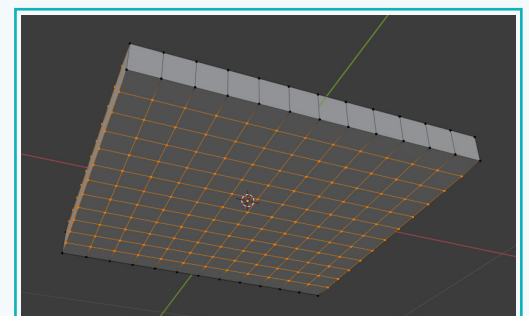
6 Press the left mouse button to confirm the number of edge loops to be added. Press the left mouse button again to confirm the location of the edge loops



7 Repeat the same process to create horizontal sections

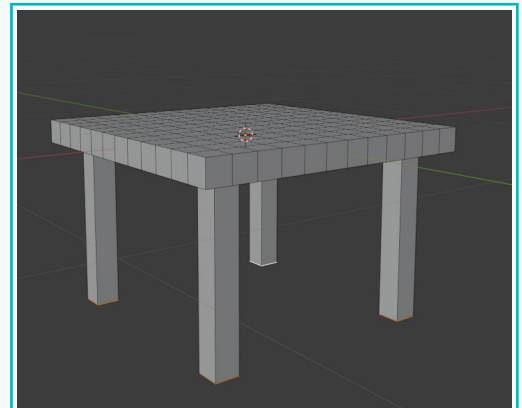
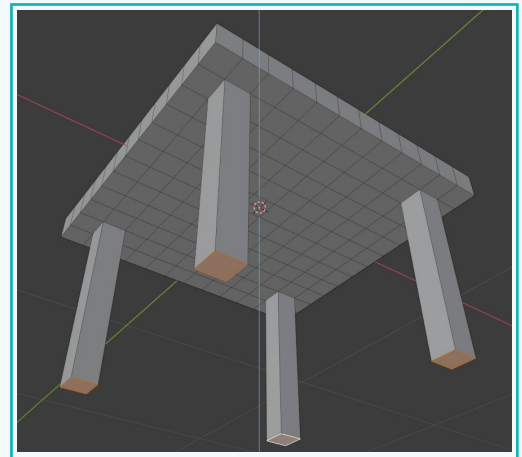
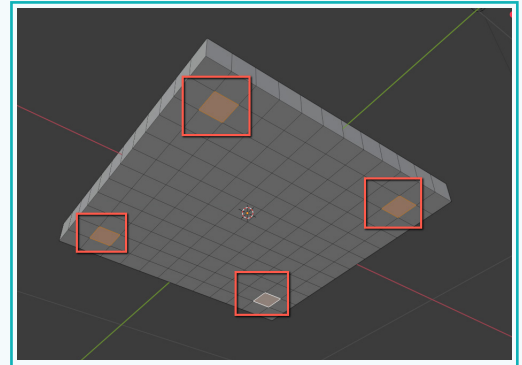
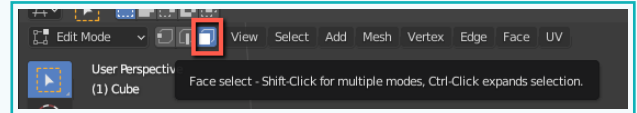


8 To create the legs of the table, we can extrude selective faces. Holding the MMB/scroll key, drag to orbit the view and see the bottom side



### Activity

- 9 To select the faces for the table legs, switch to edge mode
- 10 To select multiple faces, hold down the shift key and click on the faces
- 11 Once the faces are selected, press the E key to extrude. Drag the mouse along the z axis to make extrusions for the table legs
- 12 Orbit around the viewport to observe the object you've created

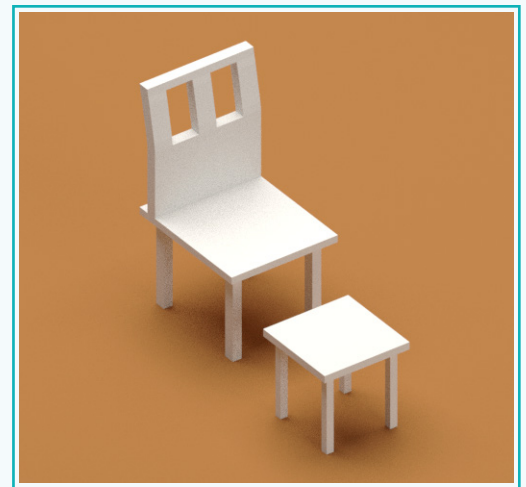


### Activity

- 13 Once the desired table is achieved, press the 'Tab' key to switch to object mode and apply materials to the object. We have added a floor to complete the scene.



- 14 Using the same technique, this table object can be modified to make a chair.



In this lesson, we learnt how to design a product in Blender using perspective view and orthographic view. In perspective view, objects and points farther away seem smaller than objects that are closer. In orthographic view, objects farther away do not seem smaller. This produces an imprecise visual output, but it more accurately depicts each object's dimensions. We also learnt about two units of measurement and how composite shapes are created in Blender.

### Exercise

#### Fill in the blanks

- 1 The default unit system used in Blender is \_\_\_\_\_.
- 2 The keyboard shortcut to Extrude is \_\_\_\_\_.
- 3 \_\_\_\_\_ is the hotkey to add edge loops to an object.
- 4 \_\_\_\_\_ mode is used to edit/modify the primitive mesh.

#### Answer the following question

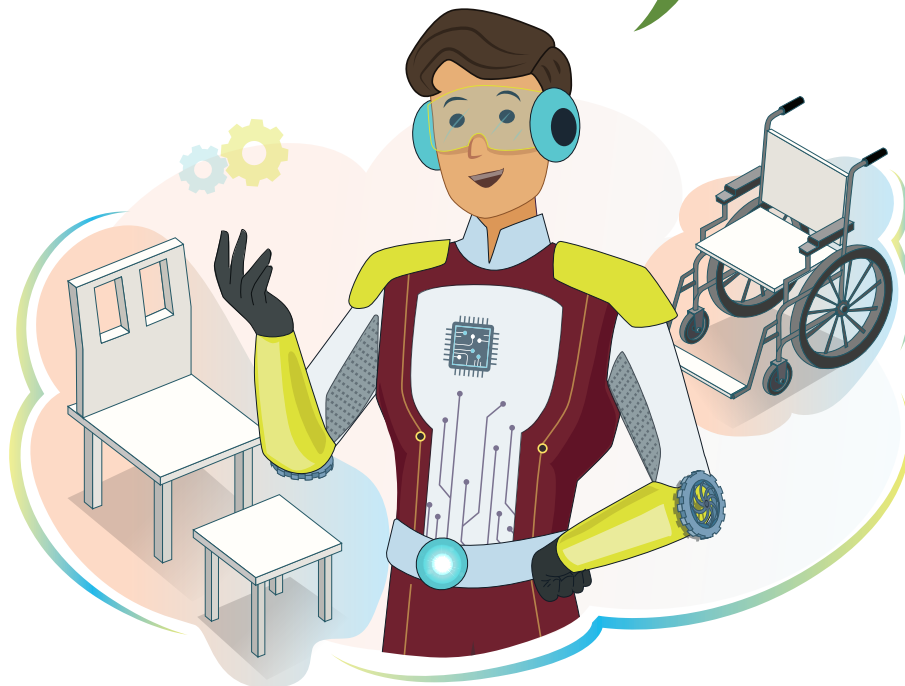
- 5 Distinguish between orthographic and perspective view.



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What other shapes can we add to this table to transform it into other objects?



### Tech Flash

- **Vertex** : A vertex is a single point or a position of a mesh.
- **Edge** : An edge is represented by a line that connects two vertices.
- **Face** : Faces are used to build the actual surface of the object.

- **Hotkeys**

|          |                                                       |
|----------|-------------------------------------------------------|
| Tab      | toggle between object mode and edit mode              |
| 7        | top view                                              |
| 5        | toggle between orthographic view and perspective view |
| 1        | front view                                            |
| 3        | side view                                             |
| 0        | camera view                                           |
| Ctrl + R | add edge loops                                        |
| E        | extrude                                               |